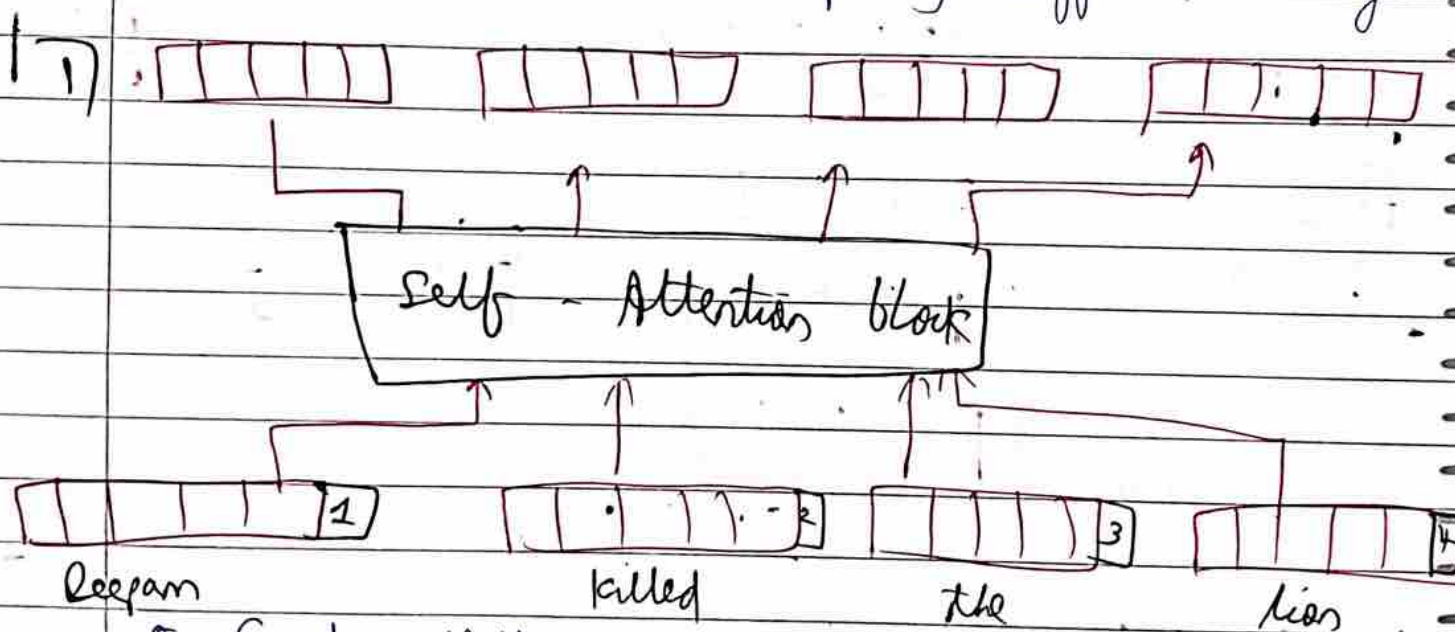


* Positional Encoding / Embedding

- Since Self-attention allowed parallel computation of generating contextual embedding vector, it has a drawback, too.

→ If we give our complete data all at once then the order of the word would lost, means sequentiality is ~~too~~ tarnished.

Eg:- Deepam killed lion → (Original Script).
lion killed Deepam → (It may be received like this)
both sentences has completely different meanings



Counting Method:-

- This is good, but we have large number of words i.e. around 1 lakh, the last position would be 100000, that's a too big number which results unbound condition.
- Also if NN is given such a big number among small it would focus optimizing big numbers only.

2) Normalize the position

- Divide each position number with the last numbered word that would range position in $(0, 1]$.
- But there's a problem with that too:-

eg:-
 $\begin{array}{ccccccc} & 0 & -1 & 2 & & & \\ \leftarrow & & & & & & \end{array}$
 Thank You $\xrightarrow{\text{pos. 2}}$ we get two different
 $\frac{1}{2}$ $\frac{2}{2}$ position number, that is
 Deepam killed the lion $\xrightarrow{\text{pos. 4}}$ not possible
 $\frac{1}{4}$ $\frac{2}{4}$ $\frac{3}{4}$ $\frac{4}{4}$

- Also problem is :- we can't get relative positioning

Deepam killed the lion
 ... 1 2 3 4
 $\swarrow$ $\downarrow$ $\downarrow$ $\swarrow$
 This is absolute position

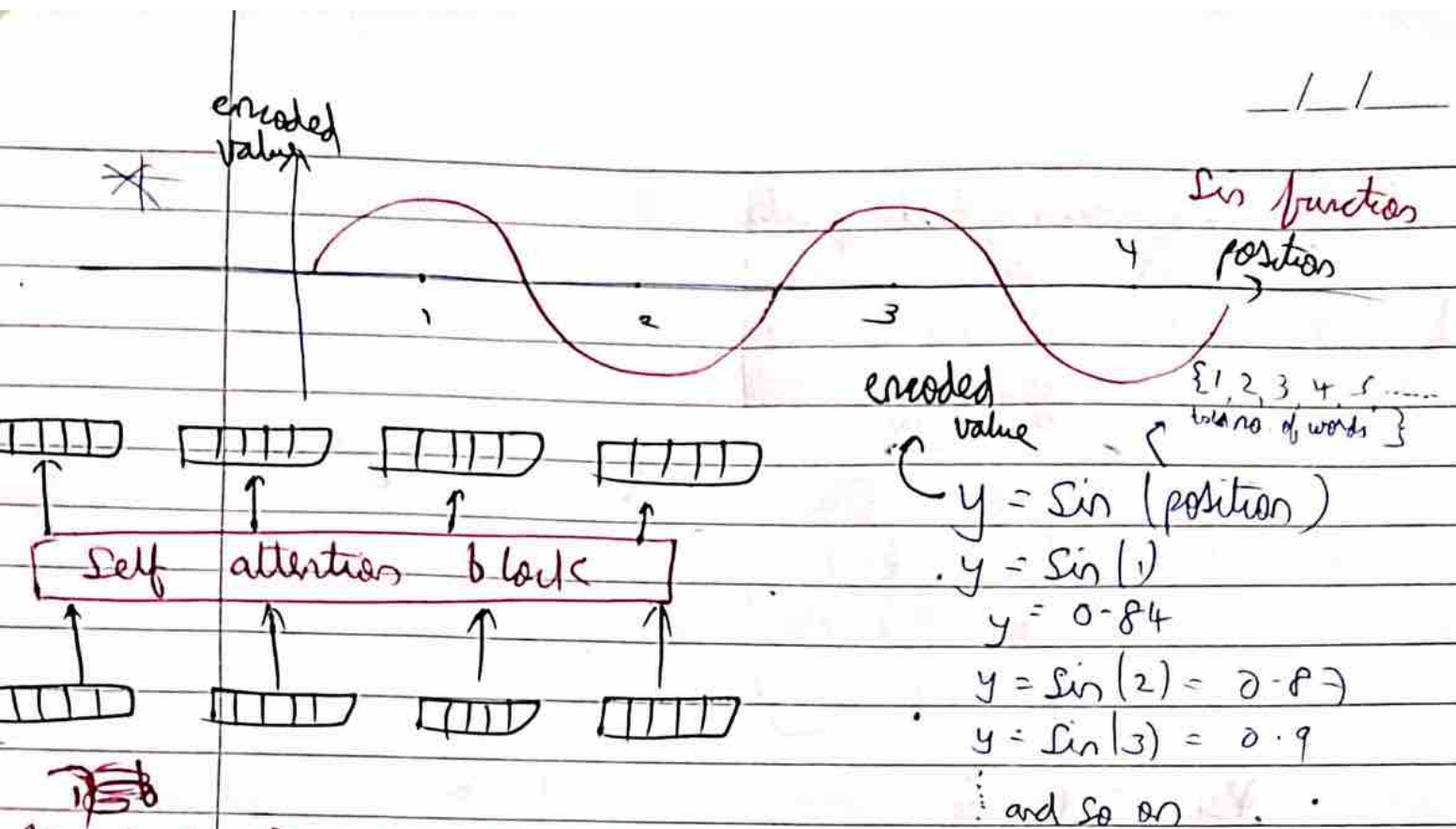
To get relative position bet the ord Deepam, we cannot do something like $3-1=2$

This problem arises when we use discrete number

* Problems

- Unbounded (bounded) - we need a function which is
- Discrete (continuous) - bounded, ~~can~~ continuous and
- Relative position (periodic) - periodic? $\rightarrow$ Answer $\downarrow$

Trigonometric
Functions!!!

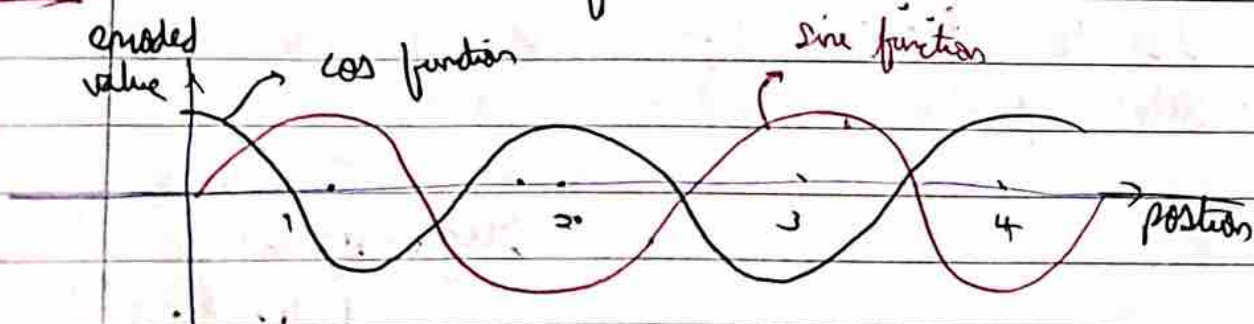


It solved :-

- 1) unbounded problem
- 2) Discrete value problem
- 3) Relative positioning $\rightarrow$ continuous graph's helps in calculating relative positions

• But there's a problem here too; different words of different position will have same encoded values

Solution :- Use cosine function too !!



$\therefore$ We could represent encoded values as 2-dim vector of cos and sin function

$$y_1 = \sin(\text{pos}) \quad y_2 = \cos(\text{pos})$$

$$\therefore y = [\sin(\text{pos}) \quad \cos(\text{pos})]$$

Deepam₁ killed₂ the₃ lion₄

$$i.e. \quad y_1 = \begin{bmatrix} \sin(1) & \cos(1) \end{bmatrix} \rightarrow \begin{bmatrix} \text{Encoded value for} \\ \text{"Deepam" word} \end{bmatrix}$$

$$y_1 = \begin{bmatrix} 0.23 & 0.89 \end{bmatrix} \quad \text{"Deepam" vector}$$

$$y_2 = \begin{bmatrix} \sin(2) & \cos(2) \end{bmatrix} \rightarrow \begin{bmatrix} \text{Encoded value for "killed"} \\ \text{word vector} \end{bmatrix}$$

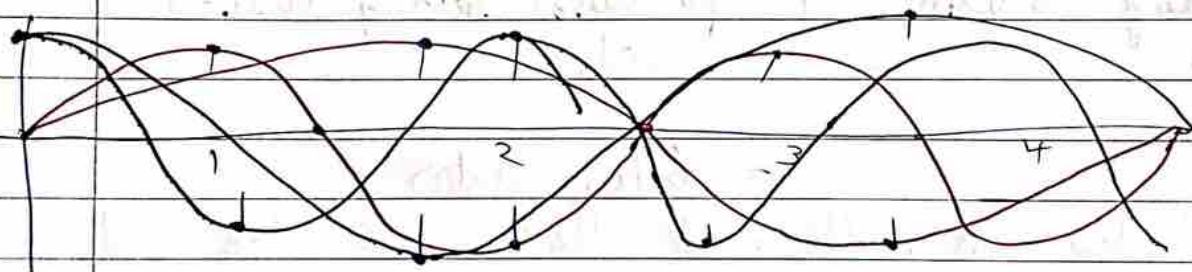
$$y_2 = \begin{bmatrix} 0.25 & 0.85 \end{bmatrix}$$

and so on

But here also, there are change that a particular positional words have same value.

Solution :- Increase or add more no. of pair of (sin cos) function with decreasing frequency

eg:-



Deepam₁ killed₂ the₃ lion₄

$$y = \sin(pos), \quad y = \cos(pos)$$

$$y = \sin(pos/2), \quad y = \cos(pos/2)$$

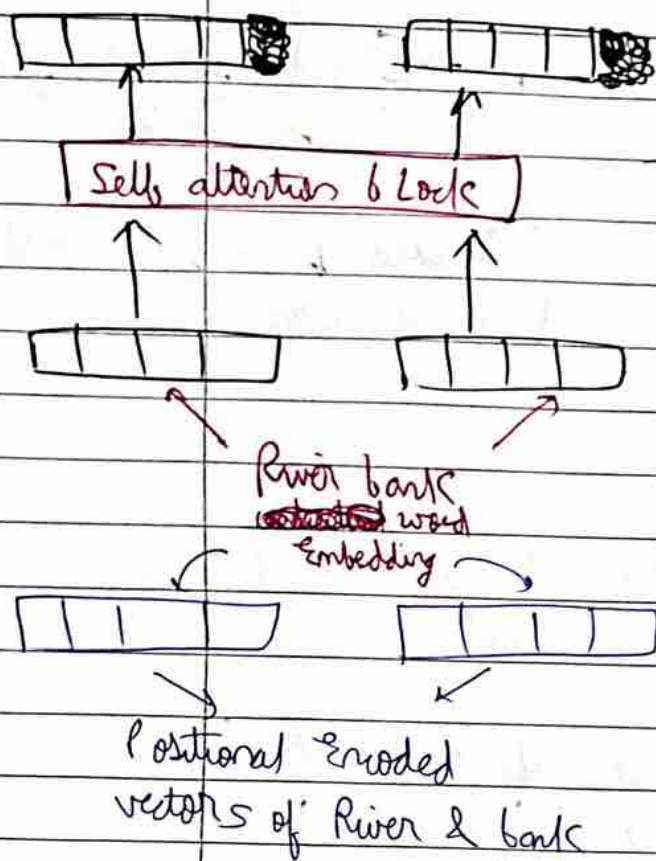
Deepam is now represented as 4-dim vec = [, , ,]

$$\therefore y_1 = \left(\sin(pos), \cos(pos), \sin\left(\frac{pos}{2}\right), \cos\left(\frac{pos}{2}\right) \right)$$

For b dim vectors $\rightarrow y = \sin(pos/3), y = \cos(pos/3)$

vector = [, , , , ,]

6-dim vector



The dimensions of a word's word embedding vector MUST BE EQUAL to the dimension of positional Encoding vector

Then what we do, we directly Add them, do additive sum



embedding (6 dim) + positional encoding (6 dim)
vector

= 6 dim vector

Why? Earlier in multi-head attention we also did concatenation, then why not here? → because concatenating it would increase the dimension i.e. $6 + 6 = 12$ dim vector, therefore it becomes computationally more complex and expensive for training.

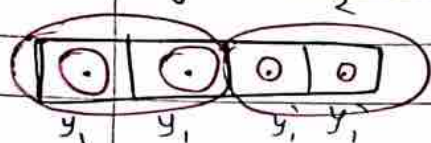
additive

So your Embedding vectors would have two information:

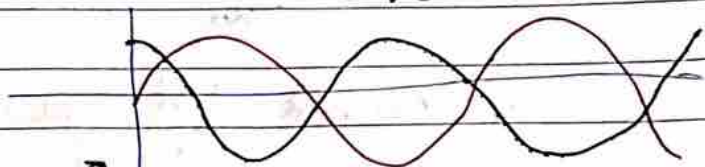
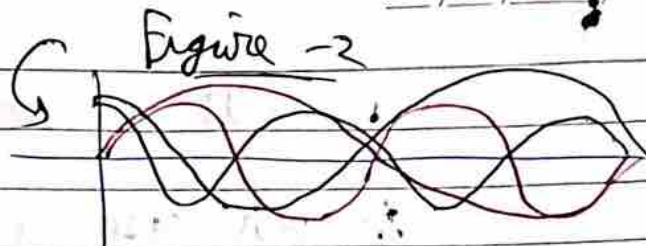
- 1) Embedding vector information
- 2) Positional encoding information

let's assume the position of river is 1

* eg:- River



position encoding vector of river



- To calculate y_1 & y_1' (1)

We need to use figure-2 which has high frequency

i.e. $y_1 = \sin(1)$, $y_1' = \cos(1)$

Figure - 1

- To calculate y_1' & y_1'' (2)

We need to use figure - 1 which has low + high frequency sin & cos function

i.e. $y_1' = \sin(1/2)$, $y_1'' = \cos(1/2)$

$$\therefore y_1 = \begin{bmatrix} \sin(1) & \cos(1) & \sin(1/2) & \cos(1/2) \end{bmatrix}$$

$$\therefore y_1 = \begin{bmatrix} 0.19 & 0.87 & 0.21 & 0.86 \end{bmatrix}$$

↳ positional encoding of "river" word

* Formula use to calculate frequency in "Attention is all you need"

$$P.E_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/(d_{model})}}\right) \dots \left(\begin{array}{l} d_{model}: \text{dimension} \\ \text{of embedding vector} \\ \text{in our } (d_{model} = 4) \\ \text{for river word above} \end{array} \right)$$

$$P.E_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/(d_{model})}}\right) \dots \left(\begin{array}{l} pos: \text{position of the} \\ \text{word} \end{array} \right)$$

~~i will range from 0 to 1~~

~~0 → use sin function~~

~~1 → use cos function → incorrect~~

i will range from $(0 \text{ to } \frac{d_{\text{model}} - 1}{2})$

In our case $i = 0 \text{ to } 1$

$$\text{Since } d_{\text{model}} = 4 \therefore \frac{4}{2} = 2$$

Watch ~~the~~ lect. again from Timestamp (55:25 to 1:04:06)
(Lec. no. 78)

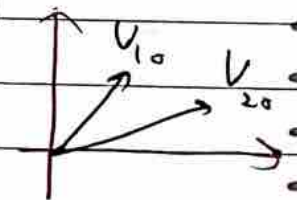
* It solved the problem of calculating relative position, how?

- i) Suppose there exist's ¹⁰²⁴ ~~1000~~ position vectors
ii) let's say we take V_{10} position vector

There exists a scalar matrix (M_1), which, when multiplied to V_{10} we can go to V_{20} 's vector position, this is linear transformation

$$\text{i.e. } V_{10} \times M_1 (\text{matrix}) (\text{linear transformation})$$

$\downarrow$
 V



If that matrix ²⁰ (M_1) is multiplied to say V_{30} i.e. $V_{30} \times M_1 \rightarrow$ we will go to V_{40} 's ~~vector~~ position

This means, this matrix (M_1) has the ability of 10 delta distance changing capability

Like that all types of deltas exist for each such type of Matrix

$$\text{Matrix } (M) = M_2, M_1, M_3, M_4, M_7, M_5, \text{ etc.}$$

i.e. $\Delta = 5$ (for M_2 matrix)

$$\therefore V_{12} \times M_2 \rightarrow V_{17} (\text{position vector})$$

* Very Important Question?

Q) How can we combine the semantic information (word embedding) and the positional information (positional encoding) into a single vector without "overwriting"?

→ Answer:-

1) When you add the embedding vector to the positional encoding ~~larger~~ vector, you're combining two different kinds of information but the values are added element-wise (i.e. each component of positional vector is added to the corresponding of the embedding vector).

2) Since this is element-wise addition, both pieces of information (semantic from embedding and positional from encoding) are blended together without losing the distinctive of either.

3) The reason positional encoding doesn't overwrite the embedding's meaning is that each dimension of the embedding vector capture some small part of the word's meaning. Positional vectors often use small magnitudes compared to the embedding vectors; meaning the positional information acts like a fine adjustment to the embeddings, instead of dominating them.